

Creative Media Analyzer

Product Requirements

An AI-assisted creative analytics tool for performance marketers

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Deployment <https://media-analyzer-theta.vercel.app>

Repository github.com/theodoreheroldrajan-sketch/media-analyzer

1. Problem statement

Performance marketers running paid campaigns on Meta and Google generate dozens to hundreds of creative variants per quarter. The platforms report metrics at the creative level (impressions, clicks, spend, conversions, revenue) but offer no structured analysis of *why* certain creatives outperform others. Practitioners fall back on manual eyeballing, ad-hoc spreadsheets, or inconsistent tagging conventions that decay over time.

The result is decision-making by anecdote. A marketer might believe "warm colors work for our brand" because the last hero ad happened to feature them, when the real driver was an unrelated copy hook. Without structured variable extraction and a sample-size-aware comparison, post-campaign learnings are unreliable and rarely transfer to the next brief.

Existing platforms address fragments of this problem: agency tools tag creatives manually (expensive and inconsistent), MMM software focuses on channel-level rather than creative-level signal, and generic LLM image taggers extract free-text descriptions that cannot be grouped or compared. None close the loop from image to structured variable to performance to ranked insight in a single workflow.

2. Target user

Primary persona - In-house performance marketers at consumer brands managing GBP 50,000-plus monthly paid spend on Meta and Google, with 50 or more creative variants per quarter. They are comfortable in ad platform dashboards but not in Python or R. They want to know which creative patterns to brief next, supported by evidence stronger than gut feel.

Secondary persona - Agency planners running paid media for several mid-sized clients. Same problem shape, larger total dataset, less continuity between campaigns.

Lighthouse user - A high-volume DTC advertiser running approximately GBP 150,000 per month in spend across roughly 200 creatives. This scale is the canonical Pro-mode case: enough data for proper modeling, technical enough to read coefficients and p-values.

3. Value proposition and differentiation

What is commodity. Vision-based extraction of structured variables from images is now a near-commodity capability available through any major LLM provider with tool-use. The tool depends on this commodity (Claude Haiku 4.5 with forced tool_use) but does not claim it as differentiation.

What is differentiated.

- **Schema rigor.** A fixed set of 24 universal variables (format, visual, color, text, brand, CTA, strategy) plus 4-6 category-specific variables drawn from one of ten brand-vertical templates. Variables are typed (boolean, enum, integer, string) so they can be grouped, compared, and eventually

regressed. The schema is project-versioned: a re-run extracts under the same definitions.

- **Lite versus Pro methodology split.** Two analysis paths gated on dataset size. Under 100 creatives runs descriptive group-by analysis with explicit sample-size confidence labels. At 100 or more creatives, the tool surfaces multiple-regression outputs (coefficients, p-values, interaction matrices). The split is honest about what each path can and cannot infer.
- **Trust score.** A composite 0-100 quality indicator shown above every dashboard. Floor-gated: the lowest of creative count, mapping coverage, and data completeness caps the score; impression volume and bucket balance contribute proportionally on top. Marketers see the floor on how seriously to take the numbers before they read them.
- **Planned personal-knowledge-base layer.** The brand context passed into extraction (brand, category, KPI, audience, campaign goal) is the seed for a markdown-based knowledge store the operator maintains over time. Repeated campaigns inherit context rather than starting from zero.

4. Scope (v1)

In scope.

- Static image creatives (PNG or JPEG), one file per variant.
- CSV exports from Meta Ads Manager and Google Ads. Auto-detected column mapping for the standard headers; the upload module accepts variants on filename, ad_id, ad_name, impressions, clicks, spend, conversions, revenue, date, campaign, ad set, platform, placement.
- Six-method matching cascade linking creatives to performance rows (exact, normalised, embedded platform ID, prefix, contains, fuzzy Levenshtein).
- Structured variable extraction via Claude Haiku 4.5 with tool_use forcing.
- Group-by analysis for any chosen metric (CTR, CPC, CPA, CVR, ROAS).
- Dashboard with trust score, key metrics, variable explorer, sortable performance table, ranked creative gallery, and an insights panel.
- CSV exports of variables, performance, and a combined view.
- Interactive demo with sample data so portfolio visitors can explore without API budget exposure.

Out of scope (v1).

- Video and dynamic creatives.
- Real-time platform API integrations (Meta or Google) - CSV uploads only.
- Multi-user collaboration, comments, or shared workspaces.
- Causal claims. The tool is correlational by design and labels outputs as directional hypotheses.

5. Lite versus Pro

The application splits its dashboard on a hard threshold of 100 creatives. The threshold is set in `src/app/api/dashboard/route.ts` and is justified in the methodology paper as the conservative floor for an OLS model with roughly 25 predictors.

Aspect	Lite (under 100)	Pro (100 or more)
Primary user	Solo marketer, small ad set, founder	In-house growth team, agency planner
Statistical method	Group-by descriptive	Group-by plus multiple regression
Visualizations	Bar chart	Bar, scatter, regression, distribution, heatmap
Mapping UI	Flat confirmed table	Match cards with confidence and method badges; suggested and u
Variables UI	Toggle list	Four-tier: Universal, Category, AI suggestions, Custom builder
Insights tone	Plain English	Coefficient and p-value referenced inline

Implementation note. The Pro UI ships in the interactive demo with mocked statistics so the experience is portfolio-visible. The production regression backend is designed in the methodology paper but is not yet wired - it will be built against the first real 100-plus creative dataset rather than synthesized in advance.

6. Key features (shipped)

Nine-step wizard: Home, Setup, Instructions, Upload, Mapping, Variables, Analysis, Dashboard, Settings.

- Image upload to Supabase Storage with PNG/JPEG validation and 10 MB cap per file.
- CSV parser (PapaParse) with auto-detected column mapping across Meta and Google export conventions; 40-plus column-name aliases handled automatically.
- Six-method matching cascade with confidence scoring per match and a manual override flow for the unmatched.
- Variable schema builder: 24 universal definitions, 10 category templates, custom variables in Pro.
- Hypothesis pre-registration: operators flag up to 5 variables they have a specific hypothesis about; the dashboard partitions analysis into a "Hypotheses tested" section and an exploratory section with Benjamini-Hochberg FDR adjustment on the exploratory p-values. Persisted to the projects table.
- Snapshot data layer for re-uploads: CSV re-imports are preserved as historical snapshots; the dashboard filters analysis to the latest snapshot and surfaces a "Showing snapshot N of M" indicator.
- Bootstrap 95% confidence intervals on every Lite delta, rendered as whisker overlays on the bar chart and numeric ranges in the performance table.

- Streaming AI extraction over Node.js runtime via NDJSON (Edge runtime is incompatible with the Anthropic SDK).
- Live cost tracking per image, priced at the Haiku 4.5 rate of USD 0.80 per million input tokens and USD 4.00 per million output tokens.
- Dashboard: metric switcher, trust score gauge (floor-gated composite with five sub-scores), variable explorer (bar in Lite; bar plus four more chart types in Pro), variable performance table with hypothesis rows pinned and exploratory ranked by noise-adjusted effect size, ranked creative gallery, insights panel.
- Settings page exposes per-project API usage observability: total tokens used, cumulative cost, run count, last-run timestamp.
- Edit project metadata flow: settings page provides an affordance to return to the setup wizard with the existing project context loaded.
- Three CSV exports: variables, performance, and a combined view.
- Project delete with cascade across all nine database tables and the storage bucket.
- Comprehensive instructions page with Meta and Google export step-by-step guides, sample CSV previews, common pitfalls, and a pre-upload checklist.
- GitHub Actions CI on every PR to main: TypeScript typecheck, ESLint, Next.js build.

7. Success criteria

Functional. A user starting with a folder of 50 creative images and a fresh Meta CSV export can complete the full pipeline (setup through dashboard) in under 10 minutes. The trust score reaches Good or better. At least three variable-value pairs return a delta with at least Medium confidence.

Decision quality. The output passes the "would a marketer change their next brief?" test - that is, the ranked insights surface at least one pattern the user did not already know, or refute one they wrongly believed.

Trust. The user can articulate, after reading the dashboard, both what the data shows and what it does not show. Sample-size labels, the regression-unlock indicator, and the trust score together make the limits legible.

8. Non-goals

These are deliberate omissions, not future work.

- **No causal claims.** Group-by deltas are descriptive. Regression coefficients (when shipped) are conditional associations. The tool produces hypotheses, not causes.

- **No real-time platform integration.** CSV uploads are the contract. Real-time APIs change too often and require credential management out of scope for v1.
- **No multi-user collaboration.** Designed for a single operator. Shared workspaces are a separate product surface.
- **No video creatives.** Frame extraction, motion tagging, and audio variables are a different problem and v2 territory.
- **No machine learning beyond OLS.** v1 stays in the regime where outputs are interpretable line by line. Random forests and gradient boosting are listed as future extensions only.

9. Future direction (post-v1)

Production regression backend. Move the mocked Pro statistics out of `demo-data.ts` and into a real server-side OLS pipeline. Gate on the first 100-plus creative real dataset to validate the design end-to-end before generalizing.

AI insight narration. Replace the current placeholder insights panel with Claude-generated explanations grounded in the actual coefficients and deltas.

Personal knowledge-base layer. Markdown notes (Obsidian-compatible) the operator maintains per brand. The notes get injected into extraction prompts so repeated campaigns inherit context.

Longitudinal analysis (v3). Cross-snapshot analysis once real multi-snapshot data is available. Use cases: creative fatigue (CTR decay within a campaign run), seasonal effects, pre/post brand-campaign comparisons, platform algorithm shifts. Methodological choices (continuous time variable, fixed effects per snapshot, panel data with creative-level random effects) depend on the empirical structure of real data. Gated on the first real user accumulating 3-plus snapshots over a 60-day window.

Full snapshot management UI. The data layer is shipped; the UI for browsing past snapshots, comparing two snapshots, and rolling back is deferred until the operator has used the tool with multiple uploads and given feedback on what's actually useful.

Additional platforms. TikTok and LinkedIn CSV formats. Demand-driven.

Optional advanced modeling. Mixed-effects models for campaign-nested data and Bayesian regression for small-sample uncertainty - explored only if real datasets justify them.

Appendix: Confirm with Theo

Items not determinable from the repository code and history. To be filled in before publishing.

- Primary persona positioning - is the Betterhalf.ai-style customer (GBP 150,000-plus monthly spend) the explicit primary persona, or one lighthouse example within a broader target?

- Naming and pricing strategy for Lite versus Pro - subscription tiers, single price, internal only?
- Personal knowledge-base layer - intended scope (markdown-only? embedded retrieval?), and rough timeline.
- Strategic direction - paid-media focus only, or expand to organic and social creatives?
- Should the live deployment URL be promoted publicly, or kept private for portfolio sharing only?